

SERJO BARRON

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EDUCATION

University of California, Santa CruzSep 2020 – Jun 2024

Bachelor of Science – Computer Engineering - **GPA: 3.70**

EXPERIENCE

Software Engineer InternOct 2025 – Feb 2026

BotStacks – San Francisco, CA

- Built reusable UI components and **dashboard analytics** features in TypeScript within a large-scale monorepo, integrating **REST APIs** to surface user-specific project metrics and improve data visibility.
- Designed and built the platform onboarding flow across **free, pro, and team** account tiers, implementing **OAuth** authentication, **Stripe** payment integration, and tier-specific user configuration in collaboration with backend engineers on PostgreSQL features.
- Centralized **50+** internal engineering documents into a **Docker**-deployed BookStack wiki, improving knowledge sharing and reducing versioning inconsistencies across teams.

Computer Science ReaderSep 2023 – Jun 2024

Baskin School of Engineering – Santa Cruz, CA

- Evaluated weekly lab reports and provided technical feedback for **200+ students** on systems-level C projects spanning abstract data structures, path-finding algorithms, Huffman coding, and cryptographic implementations.

Tech ManagerSep 2023 – Jun 2024

Slug Anime and Manga Association – Santa Cruz, CA

- Managed AV, streaming, and real-time troubleshooting for weekly events and SlugCon, supporting **1,000+ attendees**.
- Coordinated technical logistics and setup plans among officers, volunteers, and event partners, providing live support and assisting with sponsorship outreach to support event growth.

PROJECTS

CaseFile — React, Next.js, Node.js, TypeScript, Redis, Groq API, CourtListener API

- Built an AI-assisted legal case analysis platform that accepts arbitrary URLs, extracts **legal case metadata**, and generates structured case overviews through a **multi-stage retrieval and LLM pipeline**.
- Designed a **retrieval and evidence synthesis pipeline** with candidate ranking and **TTL-based Redis caching** to reduce redundant LLM inference and API requests while handling noisy speech-to-text transcripts.

Chromatica — React, Next.js, Node.js, Prisma & PostgreSQL, Spotify API

- Integrated **Spotify OAuth** to fetch personalized listening data — top tracks, artists, and configurable time windows — and extracted dominant colors from album art using **node-vibrant** to drive dynamic UI theming.
- Rendered extracted palettes as gradients and animated floating bubble backgrounds, translating raw API data into a fully reactive visual experience.

SmartMirror — React/React Native, Node.js, Electron, Python, BLE, Raspberry Pi

- Contributed to a cross-functional team of **6 engineers** to design and deploy an IoT smart mirror with a companion React Native mobile app built for **cross-platform** compatibility.
- Owned **BLE integration** end-to-end — authored the Python BLE script handling all device I/O, conducted hardware research, and troubleshooted sensor and backend connectivity between the Raspberry Pi and mobile client, achieving **sub-second synchronization**.

LoFi Scape — React, Next.js, Youtube Player API

- Built a browser-based ambient mixing tool allowing users to blend multiple ambient soundscapes alongside lofi music streamed through the **YouTube iFrame Player API**.

SKILLS

Languages	Python, C, C++, JavaScript, TypeScript, HTML/CSS
Frontend	React, Next.js, Tailwind CSS, Vite
Backend & Databases	Node.js, Express.js, SQL, PostgreSQL, Redis, Prisma, REST APIs, LLM APIs
Tools	Git, Docker, Linux, Vercel, Figma, LaTeX